Rodolfo Ostilla Mónico

Associate Professor (Profesor Titular), Fluid Mechanics

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Summary

Fluid dynamicist specializing in the direct numerical simulation of turbulent flows.

Education

Doctor of Philosophy (PhD) — **Twente University**, Physics of Fluids — Enschede, NL — Jan 2011 – Jan 2015
- Thesis: Highly turbulent Taylor-Couette flow: direct numerical simulations
- Advisors: D. Lohse (director), R. Verzicco (co-director)

Aerospace Dynamics MSc — **Cranfield University**, Aerodynamics — Cranfield, UK — Jan 2009 – Jan 2010
- Thesis: Aerodynamic flow control through weak plasma interaction

Ingeniero Aeronáutico (5-year degree) — **Universidad de Sevilla**, Aerospace Engineer — Sevilla, Spain — Jan 2005 – Jan 2010

Grado en Filosofía (Philosophy BSc) — **Universidad Nacional de Educación a Distancia (UNED)**, Philosophy — Spain — Jan 2006 – Jan 2022

Experience

Universidad de Cádiz, Assistant Professor / Profesor Titular — Puerto Real, Spain — July 2026 – present — 4 months
- Department of Mechanical Engineering and Industrial Design, School of Engineering

Universidad de Cádiz, Emergia Postdoctoral Fellow — Puerto Real, Spain — Mar 2022 – July 2026 — 4 years 5 months
- Department of Mechanical Engineering and Industrial Design, School of Engineering

Dept. of Mechanical Engineering, University of Houston, Assistant Professor — Houston, USA — Oct 2017 – Feb 2021 — 3 years 5 months
- Taught undergraduate and graduate courses in fluid mechanics and non-linear dynamics
- Supervised 5 MSc theses, 4 undergraduate research projects, and 1 PhD thesis

Harvard University, Postdoctoral Researcher — Cambridge, MA, USA; Oct 2015 – Sept 2017; 2 years

European Space Astronomy Centre (ESAC), Trainee — Madrid, Spain; Oct 2010 – Mar 2011; 6 months

Departamento de Mecánica de Fluidos, Universidad de Sevilla, Research Assistant — Seville, Spain; Jan 2008 – Jan 2009; 1 year 1 month

Projects

Aeroacústica e hidrodinámica de los instrumentos musicales tipo chimenea — Sept 2023 – Aug 2027
University of Cádiz, competitive research project funded by the Spanish State Research Agency (Proyectos de Generación de Conocimiento 2022). Main PI.
- Total funding: €81,250

CDS&E: Multi-scale Coherent Structure Extraction and Tracking for Modern CFD Data Analysis — Sept 2021 – Aug 2024
University of Houston, competitive research project funded by the U.S. National Science Foundation. Co-PI with Guoning Chen and Di Yang.
- Total funding: $453,000

Particulate Rayleigh-Bénard Flow — Sept 2021 – Aug 2024
University of Houston, competitive research project funded by the U.S. National Science Foundation. Principal Investigator.
- Total funding: $209,666

Controlling Secondary Flows by the Use of Non-Wetting Surfaces — Sept 2019 – Aug 2022
University of Houston, competitive research project funded by the U.S. National Science Foundation. Co-PI with Kamran Alba.
- Total funding: $340,000

Awards

Emergia Fellowship — Jan 2022
Fellowship funding return to Spain and research at the Universidad de Cádiz (€85,000).

Publications

Elastic turbulence in space
Rodolfo Ostilla-Mónico
doi.org/10.1017/jfm.2024.1214

Fluid dynamical pathways of airborne transmission while waiting in a line
Ruixi Lou, Milo Van Mooy, Gabriel A. Tarditti, Rodolfo Ostilla-Mónico, Varghese Mathai
doi.org/10.1126/sciadv.adw0985

Transition between boundary-limited scaling and mixing-length scaling of turbulent transport in internally heated convection
Sina Kazemi, Rodolfo Ostilla-Mónico, David Goluskin
doi.org/10.1103/PhysRevLett.129.024501

Controlling secondary flows in Taylor-Couette flow using axially spaced superhydrophobic surfaces
Vignesh Jeganathan, Tala Shannak, Kamran Alba, Rodolfo Ostilla-Mónico
doi.org/10.1017/jfm.2023.606

Instability and disintegration of vortex rings during head-on collisions and wall interactions
Aakash Mishra, Alain Pumir, Rodolfo Ostilla-Mónico
doi.org/10.1103/PhysRevFluids.6.104702

Turbulence generation through an iterative cascade of the elliptical instability
Ryan McKeown, Rodolfo Ostilla-Mónico, Alain Pumir, Michael P. Brenner, Shmuel M. Rubinstein
doi.org/10.1126/sciadv.aaz2717

Physical and geometric constraints shape the labyrinth-like nasal cavity
David Zwicker, Rodolfo Ostilla-Mónico, Daniel E. Lieberman, Michael P. Brenner
doi.org/10.1073/pnas.1714795115

Logarithmic mean temperature profiles and their connection to plume emissions in turbulent Rayleigh-Bénard convection
Erwin P. van der Poel, Rodolfo Ostilla-Mónico, Roberto Verzicco, Siegfried Grossmann, Detlef Lohse
doi.org/10.1103/PhysRevLett.115.154501

A pencil distributed finite difference code for strongly turbulent wall-bounded flows
Erwin P. van der Poel, Rodolfo Ostilla-Mónico, John Donners, Roberto Verzicco
doi.org/10.1016/j.compfluid.2015.04.007

Optimal Taylor-Couette flow: direct numerical simulations
Rodolfo Ostilla, Richard J. A. M. Stevens, Siegfried Grossmann, Roberto Verzicco, Detlef Lohse
doi.org/10.1017/jfm.2012.596

Skills

Computational Fluid Dynamics

Applied Fluid Mechanics

Interests

Philosophy of Science

Languages

Spanish
Native speaker

English
Fluent

Mandarin Chinese
B1